Md Zesun Ahmed Mia

R&D Engineer | Next-Generation Neuromorphic & Brain-Inspired Systems

Research engineer dedicated to building the next generation of brain-inspired computing hardware. Expertise in neuromorphic architectures, in-memory computing, and spintronic device integration.

Education

08/2022–05/2027 (Expected) **Ph.D. in Electrical Engineering**, *Pennsylvania State University*, University Park, PA, USA, *CGPA: 4.00/4.00*
Advisor: Dr. Abhronil Sengupta

08/2022–08/2024 **M.S. in Electrical Engineering**, *Pennsylvania State University*, University Park, PA, USA, *CGPA: 4.00/4.00* | Thesis: Neuromorphic Computing for Lifelong Learning

02/2015–04/2019 **B.Sc. in Electrical and Electronic Engineering**, *Bangladesh University of Engineering & Technology (BUET)*, Dhaka, Bangladesh, *CGPA: 3.81/4.00*

Research Experience

08/2022–Present **Graduate Research Assistant**, *Penn State*, University Park, PA

- Architecting a neuromorphic accelerator for Transformer workloads with in-situ learning capability, enabling on-chip weight updates without off-chip memory access.
- Developed brain-inspired algorithms including Equilibrium Propagation, Astromorphic Transformer, and RMAAT, achieving superior energy efficiency for edge deployment.
- Pioneered spintronic device integration using hardware-aware training, co-optimizing MTJ device physics with SNN/DNN architectures for on-chip learning.
- Engineered physics-based device models using TCAD (Silvaco/Sentaurus) and compact Verilog-A/MATLAB models from characterization data for circuit-level simulation.
- Designed energy-efficient SNN architectures targeting neuromorphic hardware, focusing on spike-based computing and local learning rules.

08/2024–05/2025 **Graduate Teaching Assistant**, *Penn State*, State College, PA

- Taught Cadence Virtuoso (schematic/layout), PDK usage, DRC/LVS, and analog/digital design flows; created hands-on lab modules and guided tool/debug workflows.
- Supervised Capstone projects for 90+ undergraduate students across communications, electronics, and firmware: supported Raspberry Pi/Arduino development, software-defined radio experiments, PCB design, and system integration end-to-end.

02/2021–08/2022 **Lecturer**, *University of Liberal Arts Bangladesh (ULAB)*, Dhaka, Bangladesh

- Taught undergraduate courses: Solid State Devices, Digital Circuit Design, Semiconductor Device Physics.

Technical Skills

Neuromorphic: SNNs, STDP, event-driven processing, brain-inspired ML, Neuromorphic hardware: Intel Loihi 2, Brainchip Akida, neuromorphic simulators (BindsNET, snnTorch, Lava)

Computational Neuroscience: Neural coding (rate/temporal/population), spiking models (Hodgkin-Huxley, LIF, stochastic LIF), synaptic plasticity (STDP, Hebbian), neuromodulation, attractor networks, predictive coding

Devices, Memory & Fabrication: Spintronic NVM (MTJ, FeFET), ferroelectric/magnetic device fabrication, SRAM/DRAM, device-circuit co-design, PIM, near-memory computing, memory-centric computing, memory controller design, thin film deposition (CVD, HDP-CVD, PVD), lithography, etching, electrical/optical/AFM/SEM characterization, magnetic probe station

HW-SW Co-Design: DTCO/STCO, Algorithm-hardware co-optimization, quantization (PTQ/QAT), pruning, mixed-precision, hardware-aware NAS, cross-layer PPA analysis, FPGA prototyping, data flow optimization, Systolic Arrays, hardware-in-the-loop training

EDA/Simulation: TCAD (Silvaco, Sentaurus), Cadence Virtuoso, Spectre, HSPICE, COMSOL, ModelSim, Synopsys, CIM (Compute-In-Memory) Systems

814-280-7244 • *zesun.ahmed@psu.edu* • *zesun33.github.io*
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Programming & ML Tools	Python, CUDA, MATLAB, Verilog, C/C++, Bash, Linux/Unix, PyTorch, TensorFlow, TensorRT, scikit-learn, Data Visualization (Matplotlib, Seaborn, Plotly), Pandas, NumPy, JMP, Jupyter, LaTeX, AI accelerator development, HPC applications, parallel computing architectures
AI Dev.	Generative and agentic AI tools (Cursor, Copilot, etc.) for research, design, and code development

Industrial Experience

05/2025–07/2025 **Graduate Technical Intern**, *Intel Corporation*, Hillsboro, OR
- Applied AI and neural network approaches to semiconductor process optimization for neuromorphic and brain-inspired computing hardware.
- Developed machine learning models for process parameter prediction supporting AI accelerator and neuromorphic device fabrication.
- Designed DOE for thin film processes enabling enhanced performance in next-generation neuromorphic computing systems.

09/2020–07/2021 **R&D Engineer**, *SEMWAVES Ltd.*, London, UK, part-time
- Delivered a 50 kW hybrid renewable system (solar + smallscale hydro) for an offgrid Bangladeshi site, supporting reliable community power.
- Owned technical leadership and coordination (vendors, field teams, stakeholders); directed device selection, integration, QA/safety procedures, and optimization against load profiles and uptime targets.

Projects

Astromorphic Transformer — Lead Student Researcher, 2022–2025. Developed a bioplausible transformer architecture leveraging neuron-astrocyte interactions to emulate self-attention mechanisms. Incorporated Hebbian and presynaptic plasticities with non-linearities and feedback, achieving superior accuracy and faster convergence on sentiment classification (IMDB), image classification (CIFAR-10), and language modeling (WikiText-2) tasks. See publication: [IEEE TCDS 2025].

Spintronic Device-Based Memory — Lead Student Researcher, 2023–Present. Led end-to-end fabrication of spintronic memory arrays using e-beam lithography, sputtering deposition, and lift-off patterning. Performed comprehensive electrical characterization (I-V, R-H loops, retention, endurance) and extracted device parameters for compact model development. Advanced non-volatile memory technologies for neuromorphic computing and in-memory processing applications.

Neuromorphic Cybersecurity with Lifelong Learning — Lead Student Researcher, 2023–2025. Developed a Hierarchical Dynamic Spiking Neural Network (D-SNN) for Network Intrusion Detection Systems combining static SNN detection (94.3% accuracy) with adaptive dynamic SNN classification using GWR-inspired structural plasticity and novel Adaptive STDP (Ad-STDP) learning. Achieved 85.3% overall accuracy on UNSW-NB15 benchmark in semi-supervised lifelong learning, demonstrating superior adaptation to new attack types while mitigating catastrophic forgetting (5.3% improvement over baseline). See publication: [arXiv], [ICONS 2025].

Select Publications

[1] **Md Zesun Ahmed Mia**, Malyaban Bal, and Abhronil Sengupta. "RMAAT: Astrocyte-Inspired Memory Compression and Replay for Efficient Long-Context Transformers". In: *International Conference on Learning Representations (ICLR)*. 2026. URL: `https://openreview.net/forum?id=sTkJdbVxsI`.

[2] **Md Zesun Ahmed Mia**, Jiahui Duan, Kai Ni, and Abhronil Sengupta. "Trilinear Compute-in-Memory Architecture for Energy-Efficient Transformer Acceleration". In: *arXiv preprint arXiv:2604.07628* (2026).

[3] **Md Zesun Ahmed Mia**, Malyaban Bal, Sen Lu, George M. Nishibuchi, Suhas Chelian, Srini Vasan, and Abhronil Sengupta. "Neuromorphic Cybersecurity with Semi-Supervised Lifelong Learning". In: *Proceedings of the International Conference on Neuromorphic Systems (ICONS '25)*. IEEE Press, 2025, pp. 235–238. DOI: `10.1109/ICONS69015.2025.00043`.

[4] **Md Zesun Ahmed Mia**, Malyaban Bal, and Abhronil Sengupta. "Delving deeper into astromorphic transformers". In: *IEEE Transactions on Cognitive and Developmental Systems* (2025).

Recognitions

- Melvin P. Bloom Memorial Outstanding Doctoral Research Award (2026)
- The Wormley Family Graduate Fellowship, Harry G. Miller Fellowships in Engineering (2025)
- Arthur Waynick Graduate Scholarship (2024),Milton and Albertha Langdon Memorial Fellowship (2023), Melvin P. Bloom Memorial Fellowship (2022)

Professional Affiliations

- Reviewer, IEEE TNNLS (2025), Design Automation Conference (DAC) (2025), IEEE MWSCAS (2025)
- Student Member, IEEE (2015-Present), Executive Member, EDS, IEEE BD (2021-2022)

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